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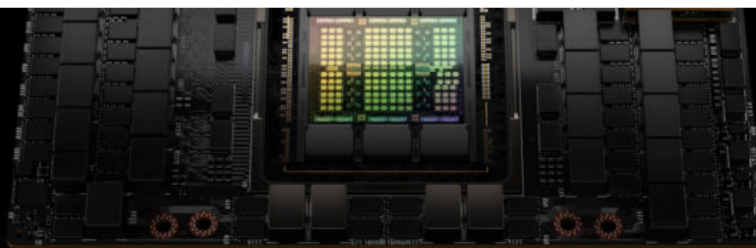
5x Faster with NV Cache

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Nov 08, 2024

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By [Amr Elmeleegy](#), [Nick Comly](#) and [Thor Johnsen](#)



AI-Generated Summary



H100 Tensor Core GPUs and 28x on the NVIDIA GH200 Superchip. In this post, we shed light on KV cache reuse techniques and best practices that can drive even further TTFT speedups.

Introduction

LLM models are rapidly advancing generation. To generate tokens, which are then used to mathematically model the understanding of the world, understanding incremental

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When reusing the KV cache, careful attention must be given to how long it remains in memory, which components to evict first when memory is full, and when it can be reused for new incoming prompts. Optimizing these factors can lead to incremental performance improvements in KV cache reuse. NVIDIA TensorRT-LLM offers three key features that specifically address these areas.

Early KV cache reuse

Traditional reuse algorithms require the entire KV cache computation to be completed before any portions of it can be reused with new user prompts. In scenarios such as enterprise chatbots, where system prompts—predefined instructions added to user queries—are essential to direct the LLM's responses in line with enterprise guidelines, this method can be inefficient.

When a surge of users interacts with the chatbot simultaneously, each user would require a separate computation of the system prompt KV cache. With TensorRT-LLM, we can instead reuse the system prompt as it is being generated in real time, enabling it to be shared across all

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Figure 1. TensorRT-LLM KV cache reuse can speed up TTFT by up to 5x

Flexible KV cache block sizing

In reuse implementations, only entire cache memory blocks can be allocated for reuse. For example, if the cache memory block size is 64 tokens and KV cache is 80 tokens, only 64 tokens will be stored for reuse, while the remaining 16 tokens will need to be recomputed. However, if the memory block size is reduced to 16 tokens, all 64 tokens can be stored across five memory blocks, eliminating the need for re-computation.

This effect is most pronounced when the input sequences are short. For long input sequences, larger blocks can be more beneficial. As is clear, the more granular the control you have over the KV cache, the better you can optimize it for your specific use case.

TensorRT-LLM provides fine-grained control over KV cache memory blocks, giving developers the ability to chop them into smaller blocks between 64 to 2 tokens. This optimizes the usage of

reducing KV cache block size from 64 tokens to 8 tokens.

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Figure 2. Impact of changing KV cache block size on inference speedup

Efficient KV cache eviction protocols

Partitioning the KV cache into smaller blocks and evicting unused ones can be effective for memory optimization, but it introduces dependency complexities. When a specific block is used to generate a response, and the result is stored as a new block, it can form a tree-like structure of dependencies.

Over time, the counters tracking the usage of the source blocks (the branches) may become stale as the dependent nodes (the leaves) are reused. Evicting the source block then requires the eviction of all dependent blocks, which would require recalculation of the KV cache for new user prompts, increasing TTFT.

more recent reuse counters. This ensures more efficient memory management while preventing unnecessary evictions of dependent blocks.

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Figure 3. A logical re

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Getting started with TensorRT-LLM KV cache reuse

Generating KV cache during inference requires a lot of compute and memory resources. Using it efficiently is critical to improving model response, accelerating inference, and increasing system throughput. TensorRT-LLM provides advanced reuse features for developers looking to further optimize TTFT response times for peak performance.

To start using TensorRT-LLM KV cache reuse check out our [GitHub documentation](#).

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About the Authors

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About Thor Johnsen

Thor Johnsen joined the NVIDIA deep learning frameworks team in 2018 and has worked on various TensorFlow and PyTorch projects, vision and language models and mlperf training. His latest work is in TRT-LLM and has focused on KV cache optimizations. Prior to joining NVIDIA, he worked on scientific computing applications for the energy industry.

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**Optimizing In
KV Cache**

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